

SUBMITTED ARTICLE



Farmworker bargaining in US agricultural labor markets

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Abstract

“Superstar firms” can be large and successful without necessarily exploiting market power over labor markets (Autor et al., *Quarterly Journal of Economics* 2020; 135 (2):645–709). In this paper, we examine this idea in an agricultural labor market setting by studying the empirical relationship between employment surplus, which is essentially the excess of a worker’s value of marginal product over their wage, and wages. We use a model of search, match, and bargaining that explains how the surplus from worker’s productivity is split between workers and employers. Our estimates show that workers’ mean productivity is \$8.67 per hour, and they receive 24.2% of employment surplus, but both exhibit substantial heterogeneity over workers. Heterogeneity in productivity and bargaining power suggests that workers who are able to generate “a bigger pie” may also earn a larger share of it. Consistent with this notion, our analysis shows a robust positive elasticity of surplus with observed wages, implying that agricultural firms gain more (surplus) by paying their workers higher wages and not necessarily through exploitation or “winner-take-all” strategy.

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KEYWORDS

agriculture, labor, monopsony, productivity, search-match-bargaining

JEL CLASSIFICATION

D43, J43, Q12

An expanding body of research examines labor market outcomes as equilibria in labor markets that are imperfectly competitive (Azar et al., 2022; Berger et al., 2022; Card, 2022; Card et al., 2021; Richards & Rutledge, 2023; Yeh et al., 2022). Imperfect competition in labor markets is commonly viewed as “worker exploitation” but, in reality, derives from heterogeneous preferences of workers regarding workplace amenities, similar in spirit to idiosyncratic consumer tastes leading to product market power in industrial organization literature (Azar et al., 2022; Bhaskar et al., 2002), rising ownership consolidation (Arnold, 2021), or frictions on the job search and matching (Berger et al., 2023; Jarosch et al., 2024; Manning, 2003). While the US agricultural sector is critical to feeding the nation, it is plagued by acute and prolonged labor shortages (AgAmerica, 2022; U.S. Department of Agriculture, 2023; Richards, 2018), which suggests that workers may also exercise a substantial degree of bargaining power in choosing to work in agriculture. Although studies on other sectors of the US economy find substantial evidence of imperfect competition in labor markets, the magnitude of this imperfect competition and its implications in farm labor markets remain unclear. Besides, given the rise of “superstar firms” (Autor et al., 2020), it is also crucial to reconsider the classical notion of “labor market power” as worker exploitation and instead focus on market imperfections from bargaining equilibria, surplus distribution, and wage-profitability link. In this paper, we examine this question and estimate the relationship between farmworker productivity-generation and wages using a sample of US farmworker data and a model of search, match, and bargaining.

Our study builds on a notion from Autor et al. (2020) that highly productive and innovative or superstar firms can have high markups and profits that do not necessarily derive from pushing wages down.¹ That is, high profitability and better worker compensation are not mutually exclusive, as efficient firms need not be the ones that suppress wages. Specifically, as Autor et al. (2020) note, superstar firms can dominate industries for reasons that reflect idiosyncratic competencies, globalization, and technological advancements but not necessarily input market power. In US agriculture, for example, evidence from USDA NASS data in 2013 shows that larger farms consistently outperform smaller operations—in one illustration, 65.2% of Large Family Farms achieve returns on assets above 5% (compared to just 4.2% of small farms).² Similarly, in the broader food system, summary evidence shows that more profitable firms actually pay higher wages (Figure 1), which is intuitive, but completely at odds with the traditional notion of large firms exercising exploitative market power over workers. We explore the possibility that farm employers in our setting can attain similar superstar status, perhaps through good management, product innovations, and decisions to grow the right crops on fertile land. We test this hypothesis by examining the relationship between equilibrium wages and earned surplus, taking into account heterogeneity in both workers' productivity and bargaining power.

Understanding the labor market status of US agricultural workers holds considerable importance. First, the agricultural sector has a chronic shortage of workers—both domestic and immigrant—due to a multitude of issues that include aging laborers, declining interest in

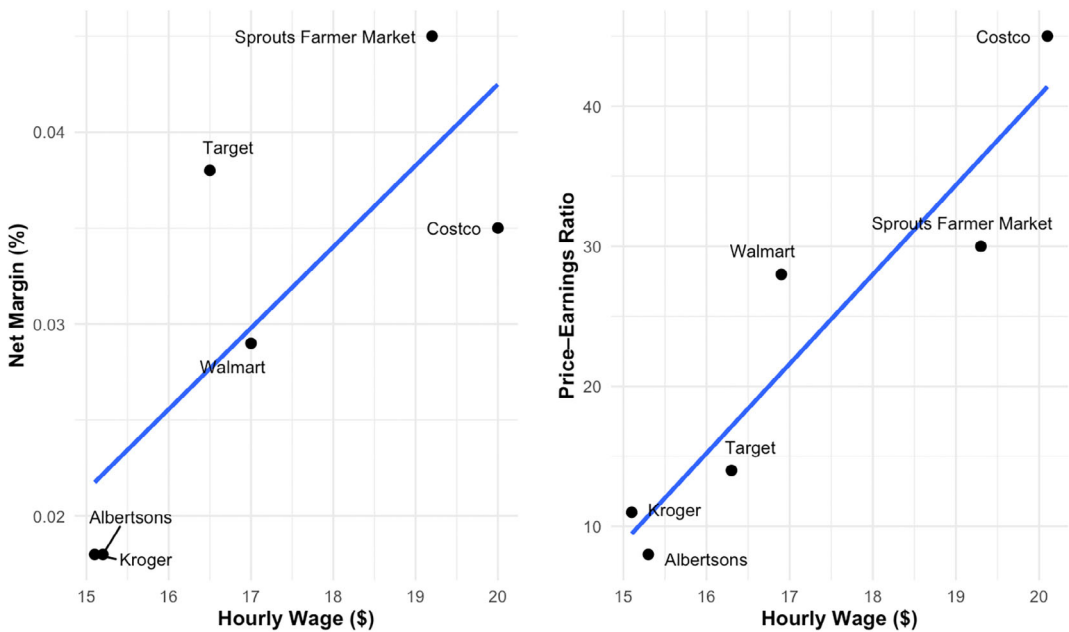


FIGURE 1 Relationship of hourly wages to price-earnings ratio and net-margins. The two panels show a positive correlation of hourly wages with employer's net margins and price-earnings ratio for a sample of US retail and wholesale firms. *Source:* The data source for price-earnings ratio is Compustat (<https://www.lseg.com/en/data-analytics/financial-data/company-data/fundamentals-data/standardized-fundamentals/sp-compustat-database>). Hourly wage data is from <https://www.indeed.com/companies/best-Retail-Wholesale-companies> and is accessed on August 10, 2024.

farming, rigid immigration policies, and increasing demands for wages (Fan et al., 2015; Richards, 2018, 2020).^{3,4} Second, from 1989 to 2022, almost 48% of the farm workers are undocumented, nearly 72% of the total workers are hired directly by growers, and close to 88% of the workers are paid on an hourly basis (Tables 1 and 2). These conditions highlight job differentiation within farm labor markets, and heterogeneity in employment terms and worker attributes forms a workforce which likely faces variations in job quality and compensation. Third, although there is an evolving literature that examines imperfect competition in labor markets in various sectors of the US economy (Arnold, 2021; Azar et al., 2022; Berger et al., 2022; Yeh et al., 2022), there is not enough parallel empirical research in agricultural settings, and it is not clear what imperfect competition means for wage outcomes for different types of agricultural workers.

Further, labor markets are at the center of an emerging policy narrative that focuses on imperfect competition, interpreted as monopsony exploitation (POTUS, 2021). However, unlike in the Executive Order on Promoting Competition (POTUS, 2021), we highlight the fact that bargaining equilibria do not need traditional monopsony explanations to yield a division of surplus between employees and employers. That is, in alignment with Autor et al. (2020), we emphasize that earned employment surplus does not mean exploitation of labor, but rather arises from competitive markets that are subject to frictions such as imperfect information, heterogeneous preferences for job-related amenities, and search costs. Our departure from the exploitation narrative of market power also builds on Sexton (2013) who argues that firms

hiring inputs (in our case, labor) in imperfectly competitive markets have no incentive to immiserate their suppliers (workers).

We measure imperfect competition in agricultural labor markets by estimating and characterizing workers' *bargaining power*, which measures the distribution of employment surplus or the gap between workers' marginal value product and the wage at which workers are indifferent between working and not working (Diamond, 1982; Flinn, 2006; Mortensen, 1982; Mortensen & Pissarides, 1994). To this end, we build and estimate a structural model of search, match, and bargaining in the Diamond–Mortensen–Pissarides (DMP) tradition which lets us directly estimate workers' bargaining power parameter $\gamma \in (0, 1)$, which is interpreted literally as the share of employment surplus that goes to workers, with the remaining share $1 - \gamma$ retained by employers. A higher value of γ indicates greater bargaining power, and wages get closer to productivity value, which is a competitive wage. In perfectly competitive labor markets, market equilibrium from forces of labor demand and supply determines wages, thus there is no room for individual bargaining or surplus to negotiate. However, under imperfect competition, search frictions—including imperfect information regarding job opportunities and worker productivity—as well as preferences over job attributes lead to wage outcomes that are typically lower than those that would emerge in perfect competition.

We extend the classical model of search, match, and bargaining in Pissarides (2000) and Flinn (2006) by accounting for several institutional features of US agricultural employment. In the model, workers and employers both search for optimal wages and arrive at a match once there is no marginal benefit of searching for either side. The resulting matches generate an employment surplus, which workers and employers negotiate through a Nash bargaining framework. In equilibrium, the balance between the employer's and worker's bargaining power, together with the worker's productivity and job preferences, determines worker's wages. In our model, however, we allow both productivity and bargaining power to depend on attributes of individual workers and of the jobs they do. In this way, our model captures worker heterogeneity and outcomes that vary across their individual employer matches.⁵

We leverage productivity and bargaining heterogeneities to study the “superstar effect” in our framework by allowing worker productivity and bargaining power, and ultimately, wages to vary by observed worker attributes like age, gender, years of experience, foreign status, methods of payment, and hiring process. Several studies document bargaining and productivity heterogeneity along some of these variables. Due to occupational segregation and work interference, for example, women are likely to have less bargaining power (Blau & Kahn, 2017). Age and experience can correlate negatively with productivity due to physical and cognitive declines from age or resistance to adapt in new working environments from more experience (Goebel & Zwick, 2009). Workers who are foreign-born are unlikely to provide higher value to their employer despite bringing diverse perspectives and skills if they do not assimilate well in the new working conditions (Lazear, 1999). In contrast, workers who are hired directly by growers can have higher productivity as they have better incentive structures and closer supervision relative to the ones hired by farm labor contractors. To test the superstar-firm hypothesis, we examine the wage-surplus relationship based on this heterogeneity of productivity and bargaining power.

We use data from the National Agricultural Workers Survey (NAWS) which is collected by the US Department of Labor and is a nationally representative survey of demographic, health, business, immigration, and workforce attributes on randomly selected US crop workers (National Agricultural Workers Survey, 2023). For this study, we use the restricted version, which contains cross-sectional information on about 25,000 workers in California from fiscal

years 1989 to 2022.⁶ Although the original dataset includes information for 48 states, we focus our analysis on California to ensure comparability across workers and avoid complexities of cross-state minimum wage variations.⁷

We find that workers earn roughly one-fourth of the employment surplus available for allocation between workers and their employers. In our preferred specification, after controlling for year, crop-type, and task-type fixed effects and worker attributes, we find that workers on average have a bargaining power parameter $\gamma = 0.242$. This means that they take 24.2% of the employment surplus, and employers take the remaining share, reflecting a notable gap in the surplus distribution and substantial power asymmetries in the agricultural labor market. Bargaining power is exogenous to the employment agreement, and depends on attributes of the employer and employee (Nash, 1951). That is, we explain how competitive wage outcomes vary over individuals according to their ability to extract their share from their employer. In the heterogeneity analysis, we find that bargaining power γ is heterogeneous across worker attributes.

In our analysis, we recognize that size of the surplus also matters besides the distribution of the surplus.⁸ While it may appear that sharing only 24.2% of the surplus implies worker exploitation, we argue otherwise by drawing on Autor et al. (2020), who highlight that large firms can be highly profitable without exploiting market power in either consumer or input markets. We examine this argument by estimating the relationship between worker surplus and equilibrium wages. We find a surplus-wage elasticity of $\epsilon = 0.162$ (t -ratio = 29.54), implying that farmers who pay higher wages also earn more surplus.⁹ Further, we find that some workers such as males, piece-rate earners, and more experienced have lower bargaining power but higher productivity than their counterparts, and hence can still benefit if productivity effects offset bargaining disadvantages. That is, we emphasize that our findings do not suggest exploitation of certain worker classes, based purely on bargaining power effects, and focus instead on how surplus relate to wages and show a potential for workers to be better-off despite having lower bargaining power.

Our study makes three main contributions. First, the paper extends the current labor monopsony literature, which examines labor market power from the lens of a differentiated-jobs approach (Azar et al., 2022; Berger et al., 2022; Card et al., 2021) and job-search frictions (Faberman et al., 2022; Jarosch et al., 2024; Richards & Rutledge, 2023).¹⁰ While these studies are industry-specific and do document the prevalence of employer power, there are no similar studies on US crop workers. Further, their insights are not directly relevant to understanding bargaining equilibria in US agriculture due to its unique features, such as a continual labor shortage and large shares of an undocumented workforce and hourly workers. Our paper fits into this monopsony literature as we re-frame the argument away from connotations of exploitation and market power that follow from the monopsony perspective, and more in terms of how real-world market outcomes reflect heterogeneity and informed business negotiations.

Second, our paper contributes to the growing literature on agricultural labor and labor shortages by highlighting the heterogeneity between workers' productivity, wages and the worker-share of employment surplus. When labor is in short supply, it is simply implausible that farmers are able to exploit workers, or even find that doing so is in their best interests. We examine the extent to which different attributes impact both workers' productivity and their share of employment surplus. In doing so, we identify the worker groups that generate more value relative to their total compensation. We argue that shifting the focus from labor market power, that is, bargaining power effects on the distribution of surplus, to the alignment between value creation and total compensation offers greater understanding of labor market inefficiencies. Most importantly, we find that the superstar-firms argument in Autor et al. (2020) also

applies to agricultural employers in that they earn more surplus by compensating their workers better and not inevitably through exploitation or a zero-sum game between employers and workers.

Our findings resonate with Efficiency Wage Theory (EWT), which suggests that firms may pay above-market wages to enhance worker productivity, lower turnover, and deter shirking, thus aligning employee incentives with firm objectives (Katz, 1986). In our analysis, the positive relationship between wages and firm surplus supports the notion that higher wages can foster improved worker performance and loyalty, consistent with EWT's predictions. However, the concept of superstar firms extends beyond this traditional scope by emphasizing labor's share of profit: although our application differs from Autor et al. (2020), the core idea is that a low labor share of value need not imply monopsony power but may instead reflect disproportionately high surplus generated through superior management practices, technological innovations, and optimal crop selections. Consequently, while EWT suggests that high wages are primarily a causal mechanism to motivate workers, superstar firms may pay high wages simply as a byproduct of their inherent productivity advantages. In this dual mechanism, whereby wages are higher with greater productivity and as a reflection of the firm's inherent efficiencies, wage premiums can arise not just from the need to motivate workers, but also from the firms' unique capabilities and market positions.

Third, our paper contributes to the discussion on how industrial policy applies to labor markets (Federal Trade Commission, 2023; Naidu & Posner, 2022; POTUS, 2021). In 2021, President Biden released "Executive Order on Promoting Competition in the American Economy" expressing concerns for unfair treatment of family farmers and their workers, importance for raising workers' power to bargain for better wages and working conditions, and to "enforce antitrust laws to combat the harmful effects of monopsony in agricultural markets" (POTUS, 2021). In this study, we test the underlying assumption of labor market power in The Executive Order in the setting of crop workers in California. Further, we offer a more nuanced view of labor market outcomes in agriculture by considering differences in wages as outcomes of negotiated equilibria, and not exploitative relationships. Our analysis examines wages through the lens of imperfectly competitive markets, and we show that unequal sharing of surplus is an outcome of bargaining equilibria, and is not necessarily related to classical notions of market power.

The rest of the paper proceeds as follows. First, we present the structural model and identification and estimation strategies. Second, we describe the data and some stylized facts about the US agriculture. Third, we illustrate the results from the structural model. Finally, we conclude.

MODEL INTUITION

Our conceptual model builds on Flinn (2006) and assumes workers and employers optimally search for jobs and workers, respectively, until a match is created, which happens when the marginal cost of search equals the marginal benefit for either side. The model assumes immediate observability of match productivity for both sides, but this should be interpreted in an "as if" sense—the observed wage and unemployment duration patterns reflect outcomes that would occur if match values were immediately known. After the match, workers and employers negotiate for wages based on their bargaining power, which is exogenous to both and depends on their negotiating skills.¹¹

Our structural approach has several advantages. First, it integrates labor market search, match, and bargaining to provide a comprehensive understanding of equilibrium wages, employment, and productivity. Second, the model is a realistic depiction of how workers and employers optimize their search efforts for employment surplus and how both negotiate or bargain for wages based on their individual and market attributes, in contrast to rigid wage-setting models in the literature.¹² Third, it provides an econometrically clean way to estimate and characterize the extent of imperfect competition in US agricultural labor markets by examining the distribution of employment surplus between employers and workers. It does this by addressing other sources of labor market imperfections like search frictions, information asymmetry, and idiosyncratic preferences for job attributes. Fourth, it generates hypotheses that are testable on our data. In particular, we are able to analyze the relationship between worker attributes and various labor market outcomes such as wages, employment surplus, and productivity.

Although on-the-job search (OJS) is an important extension of our search, match, and bargaining framework following (Flinn, 2006), our data do not track alternative offers or job-to-job transitions, which prevents us from incorporating OJS directly. For this reason, we abstract away from OJS in our analysis. Prior studies (e.g., Dey and Flinn (2005); Cahuc et al. (2006); Flinn and Mabli (2008); Gottfries and Jarosch (2023)) using matched employer–employee datasets show that incorporating OJS primarily affects the interpretation of the bargaining parameter—specifically by reducing it relative to the no-OJS case—and shifts match values upward. By contrast, these changes appear to have only limited effects on core welfare implications, because workers can trigger “bidding wars” between employers that transfer surplus to workers over time through wage renegotiation. In other words, the literature concludes that while OJS can alter both the distribution of match values and bargaining outcomes, the main policy conclusions regarding minimum wages remain robust.

If workers and employers search optimally, then our model implies that employers will earn some surplus, defined as the difference between the marginal value product of the worker and his or her “threshold wage” that would otherwise keep them at home, and workers with higher productivity will earn higher wages, depending on their ability to extract surplus from their employer. This framework generates a structural econometric model of labor market outcomes that we describe in more detail in Appendix A.

DATA

In this section, we describe our data sources, focusing on the National Agricultural Workers Survey (NAWS) conducted by the US Department of Labor, and stylized facts about crop workers in California. NAWS is highly appropriate for this study as it is a comprehensive and representative dataset on US crop workers across multiple dimensions that include demographics, employment characteristics, and income. The NAWS data follows a multi-stage sampling procedure, ensuring that it reflects the broader population of crop workers and makes it valuable to study labor market dynamics in the agricultural sector.

Although the data covers crop workers across the United States, we focus on workers in California. There are two main reasons for this. First, different states implement minimum wages in agriculture differently, and analyzing the effects of cross-state minimum wage is very challenging. The second reason is that focusing on California involves the analysis of workers in similar industries and employers that sell in similar markets.

We also illustrate key stylized facts about the agricultural labor market in California, examining trends in workers' age and average wage, types of crops and tasks, and how factors like gender, immigration status, hiring process, and payment methods are related to workers' real hourly wages. By showing these stylized facts, we aim to emphasize important patterns and differences within farmworkers in California. Our fundamental goal, however, is to understand how these labor market factors influence the allocation of employment surplus between workers and employers.

Data sources and summary statistics

We use two datasets in our analysis: the National Agricultural Workers Survey (NAWS) and minimum wage series for the period 1989–2022. The NAWS data is from the US Department of Labor and contains information on demographic, health, business, immigration, and workforce attributes of randomly selected US crop workers. The NAWS employs a multi-stage sampling technique to interview possibly different workers three seasons or cycles (spring, summer, and autumn) each year from about 90 county clusters that are selected based on the intensity of farm activity and the county's share in a season's total payroll (Fan et al., 2015; Kandilov & Kandilov, 2010; Li & Reimer, 2021; National Agricultural Workers Survey, 2023). The sampling procedure is described in detail in (National Agricultural Workers Survey, 2023) and involves randomization in each of the following levels: cycle, region, single counties or county clusters called farm labor areas (FLA), county, ZIP code, employer, and crop workers. For this study, we use the restricted version containing cross-sectional information on about 25,000 crop workers in California.¹³

NAWS provides details about farm workers along multiple dimensions—demographics, job attributes, housing, income—but our study employs only the first two categories. For demographic variables, we observe workers' place of birth, race, age, ethnicity, marital status, gender, work authorization, marital status, and years of education. Second, NAWS describes features of agricultural jobs such as crop and job types, hiring process (that is whether an employee works under Farm Labor Contractor (FLC) or grower), wages, and working hours. We summarize a list of the NAWS variables we use in our estimation in Table 1.

TABLE 1 Summary of NAWS variables used in estimation.

Variables	N	Mean	St. dev.	Min	Max
Hours per week	24,827	44.5	11.7	1	120
Age	24,827	36.5	12.7	14	88
Years of education	24,827	7.5	3.5	1	21
Years of farm work experience	24,827	14.1	11.5	0	78
Weeks worked in last 52 weeks	24,827	7.5	9.8	0.0	52.0
Real hourly wage	24,827	11.2	3.2	2.7	44.8
Real minimum wage	24,827	9.3	1.4	7.0	13.4

Note: The table indicates a summary of statistics of crop workers in the NAWS sample period 1989–2022 for demographics and workforce variables.

Source: The data source is NAWS, US Department of Labor (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

NAWS specifically surveys workers in crop-related jobs in establishments that fall under two NAICS codes (Gold et al., 2019). First, it targets workers in NAICS 111 or Crop Production, which includes enterprises such as farms, orchards, groves, greenhouses, and nurseries primarily involved in cultivating crops, plants, vines, or trees, along with their seeds. In our sample, farmworkers are involved in industries such as corn, wheat, barley, oats, and rice (Figure 4). Second, NAWS interviews workers in NAICS code 1151 or Support Activities for Crop Production, which includes establishments offering assistance for crop growth such as labor supply, aerial spraying, cotton ginning, cultivation services, farm management services, crop planting, and vineyard cultivation services. Our sample of NAWS workers in California performs duties in various aspects of crop production and related tasks such as arrangement, cleaning, harvesting, machinery operation, pest control, and crop packing (Figure 4). In our analysis, we include fixed effects across task and crop types to account for the fact that wages are likely to reflect differences in job responsibilities and crop industries.

Our model accounts for the importance of minimum wages in agricultural employment, and how minimum wage laws constrain negotiated wage outcomes. In our study period 1989–2022, we find that California's minimum wage policy impacts wage distribution for a substantial portion of the agricultural workforce on at least two grounds. First, 24.1% of the crop workers in California earn within 1% of the state minimum wage adjusted in real terms. Second, over the 33-year study horizon, California adjusted minimum wage policy on 16 instances (with the first in 1989) when considered in nominal terms, and it changed every year in real terms. Because minimum wage is binding in our data, it imposes a lower limit on the set of wages employers are required to pay. That is, the distribution of wages after a binding minimum wage is in place is different from wage distribution without a binding minimum wage. Thus, our study introduces minimum wage data, which we collect from Vaghul and Zipperer (2022). It contains information on yearly changes of minimum wages in California for 1989–2022. The second panel on Figure 3 plots the evolution of state minimum wages, rising from \$9 per hour to \$13 per hour (both in real terms) in our study period.

In our analysis, we characterize crop workers based on two variables: hourly wage rate for employed and duration of unemployment in the past year for unemployed. This helps to bring our theoretical model to data as we use this information to build a likelihood function for workers who are unemployed, who earn the minimum wage, and those earning above the minimum wage. We measure workers' wages and state minimum wages to real terms by deflating them with an appropriate price index. We measure the (un)employment duration using the “weeks worked” variable, which reports the number of weeks each worker was working in the previous 52 weeks. We also use the fact that the NAWS data is a repeated cross-sectional survey, and that it provides retrospective account of workers' employment records over the past year.

We summarize our NAWS data in Table 1 and highlight a few important facts about farmworkers in California. First, the second panel in Figure 2 shows the average age of crop workers is rising over time and has reached 44 years in 2022, up from 34 in 1989. Over 50% of the workers are over 43 years of age in 2022. An aging workforce indicates more farm experience and likely higher productivity, but it can also signal “lock-in effect” where workers find it difficult and costly to switch jobs, potentially leading to reduced bargaining power (Cahuc et al., 2006). We seek to test this hypothesis of the lock-in effect in our analysis. Second, Figure 3 shows a dense concentration of real hourly wages at \$9.6 and a small portion of the workforce earns over \$20 per hour. The second panel in the same figure illustrates that both real hourly wages and state minimum wages trend upwards, at least after 2013. Upward-trending real wages suggest that workers are earning more of the employment surplus from

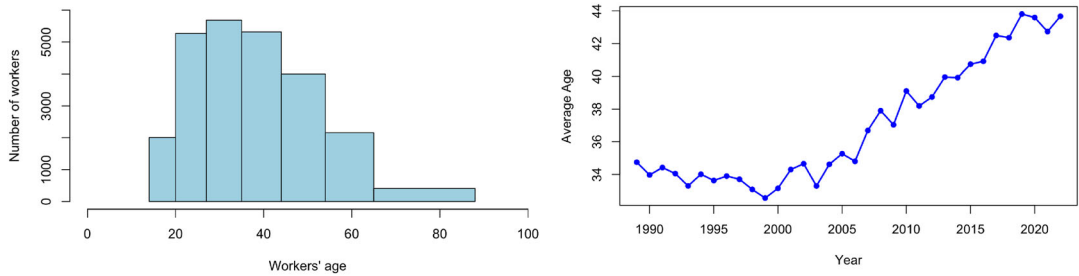


FIGURE 2 Age distribution and trends in the average age of US crop workers. The first panel shows the distribution of workers' age in NAWS data for the period 1989–2022. The median age is 35, and nearly 40% of workers are above the age of 40. The second panel shows the evolution of the average age of crop workers across years. Their average age in 2022 is 44 years, up from 34 years in 1989. *Source:* The data source is NAWS, US Department of Labor (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

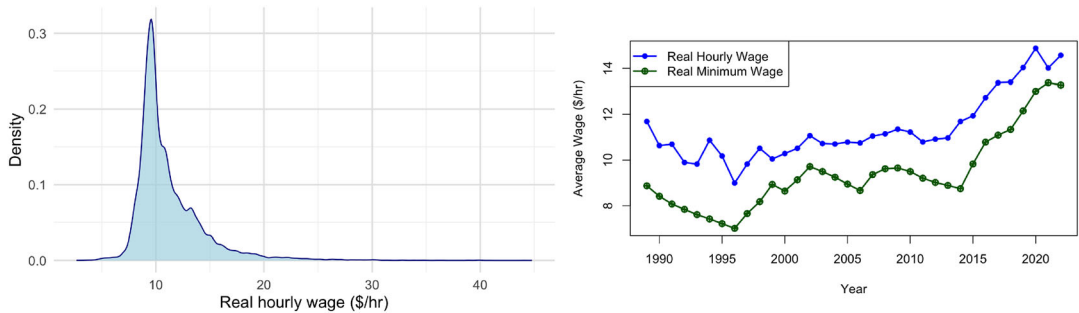


FIGURE 3 Wage distribution and trends in the average wage. The first panel shows a kernel density plot of workers' real hourly wage in NAWS data for the period 1989–2022. The plot peaks at an hourly wage of \$9.59, indicating wages are densely concentrated around this rate. The second panel shows the evolution of the average hourly wage of crop workers and state minimum wage across years. The two wage types move nearly parallel and start to rise after 2013. *Source:* The data source for real hourly wages is NAWS, US Department of Labor (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>). Real minimum wage data is from <https://github.com/benzipperer/historicalminwage/releases>.

firms, but it may also be the case that firm profitability (and hence marginal value products) are rising. Our empirical model disentangles these two explanations for rising wages.

In Table 2, we present demographic and employment data on crop workers, and highlight heterogeneity in their composition. We notice a workforce that is mainly male (80.5%) and foreign-born (94.6%), with a relatively similar distribution between documented (52.3%) and undocumented (47.7%) workers. Most workers are hired directly by growers (71.8%), while a substantial fraction works for Farm Labor Contractors (FLCs) (28.2%). The majority receive hourly wages (88.3%), with a minor share paid on a piece-rate basis (11.7%). Each of these features of our data are likely important to both the productivity of workers and how earned surplus is allocated between firms and workers, which we address in more detail below.

We show the distribution of jobs and crop categories that US crop workers are involved in the period 1989–2022 in Figure 4. From the first panel, we see that the most common tasks are semi-skilled, and in pre-harvesting, harvesting, and post-harvesting, while supervision is the least common task. The second panel indicates that the majority of workers work in fruits, nuts,

TABLE 2 Frequency distribution for NAWS binary variables.

Binary variable	Category and share of total workforce	
Gender	Male	Female
	80.5%	19.5%
Employer type	Grower	Contractor
	71.8%	28.2%
Foreign status	Native	Foreign
	5.4%	94.6%
Wage payment method	Hourly	Piece rate
	88.3%	11.7%
Status	Documented	Undocumented
	52.3%	47.7%

Note: Share of workers for each class of binary variables in the NAWS data for the period 1989–2022.
Source: The data source is NAWS, US Department of Labor (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

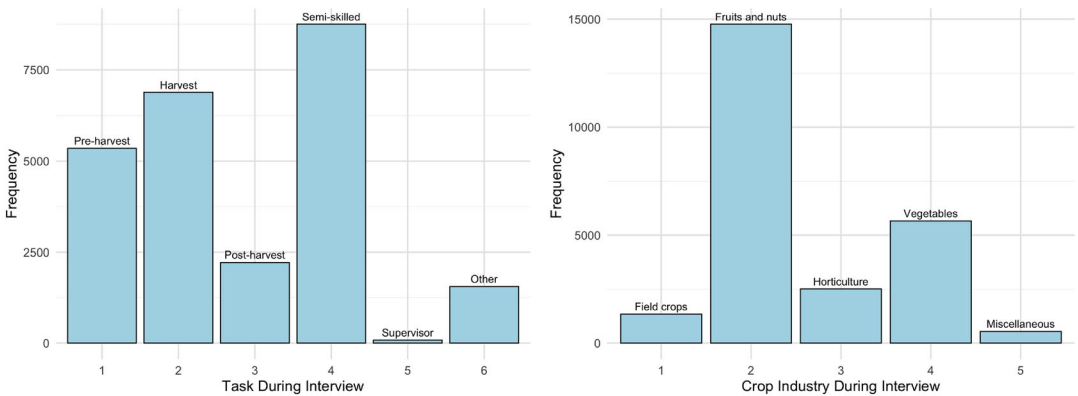


FIGURE 4 Types of task and crop during interview of US crop workers. The first panel shows frequencies of types of tasks that workers in our sample are involved in during the period 1989–2022. The second panel shows types of crop industries they work in during the interview. Source: The data source is NAWS, US Department of Labor (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

and vegetable farms, while field crops and horticulture are also important crop industries. This concentration in particular types of crops and tasks can impact labor market dynamics as these activities are seasonal and labor-intensive in nature. We account for differences in wages across jobs and crops in our empirical model below as there is clearly substantial heterogeneity in this dimension in our data.

Our study explores how different groups of workers obtain different levels of compensation based on the relative strength of their productivity level and bargaining power. As a first step in that direction, we examine the heterogeneity of how real hourly wages vary across key worker and employer attributes. The five panels in Figure 5 demonstrate the evolution in real hourly wages across various worker groups—gender, employer type, foreign status, documented status, and payment method. While wages have generally increased during 1989–2022, the degree of

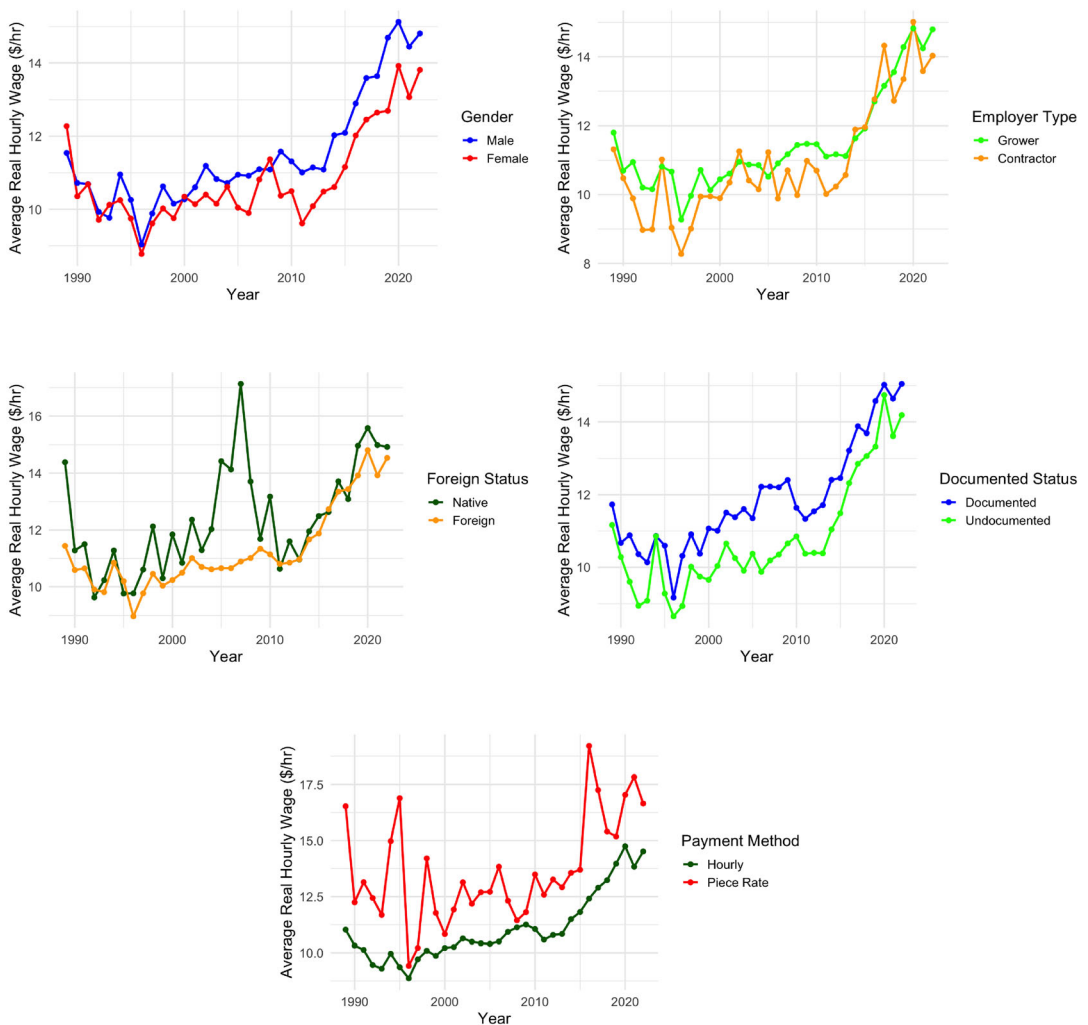


FIGURE 5 Real hourly wages across years and by worker groups. The five panels show the evolution of average hourly wages across years on the basis of five different variables for the period 1989–2022. *Source:* The data source is NAWS, US Department of Labor (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

increase differs among different groups. Male workers and those hired by FLCs have higher wages compared to their counterparts. Native, documented, and hourly-earning workers experience consistently higher wage growth than their counterparts. However, wages are only indirect measures of the value of workers to their employers in a setting with search frictions and other market imperfections. It is possible for some of these groups to have higher productivity but lower bargaining power, and the relative effects of the two forces can influence the ultimate wage determination.

In summary, the data suggest significant changes in the composition and wage distribution of California farmworkers, highlighting heterogeneity in demographics and employment characteristics. These trends imply an intricate interplay among the labor market environment, worker productivity, bargaining power, and labor market outcomes. The patterns that we observe in our data necessitate a deeper investigation into how employment conditions define

labor market outcomes, particularly the division of surplus across various worker groups. To perform a rigorous analysis of factors determining the surplus distribution and to have a nuanced understanding of agricultural labor markets, we develop an empirical model that assesses how worker and firm attributes interact with wage outcomes.

Empirical analysis

In this section, we describe how we bring our theoretical model of search, match, and bargaining (Appendix A) to the data. We discuss the mathematical derivation of our empirical model in Appendix B and only describe our strategy here. Given that the NAWS data is repeated cross-sectional and not matched employer–employee data, which is ideal for estimation, our study follows others in the literature (Flinn, 2006; Richards & Rutledge, 2023) to estimate the model parameters. The approach involves building a likelihood function that uses individual-level observations of real hourly wages and unemployment duration to evaluate model parameters such as the job arrival rate, job separation rate, bargaining power, and productivity distribution.

The likelihood function forms a core part of our empirical strategy, and we build it by classifying workers into three distinct groups: those who are unemployed (and hence “earning” less than state minimum wage), those who earn exactly the minimum wage, and those who earn wages above the minimum wage. We calculate the likelihood of group membership using the probability of specific unemployment durations within the population. That is, each of the three segments of the labor market has a “likelihood contribution,” and we combine the contributions of all three groups into a single likelihood function for estimation purposes.

For unemployed workers, the likelihood contribution assumes that the duration of unemployment follows a negative exponential distribution, which simplifies the empirical analysis by associating the gap between job offers with a Poisson process and making it mathematically tractable. The unemployment distribution is based on a hazard rate, which indicates the chance that a proposed job pays below the minimum wage. The likelihood of finding a given unemployment spell then results from this hazard rate, accounting for the rate at which unemployed workers get job offers.

The likelihood contribution from workers earning the minimum wage accounts for the probability that a worker who receives a job offer is constrained by the minimum wage. Specifically, we integrate the minimum wage constraint into the Nash bargaining framework, which is used to model surplus distribution. Intuitively, this reflects the probability that their negotiated wage, which would otherwise be lower based solely on their productivity and bargaining power, is now raised above the mandated minimum wage.

For workers earning higher than the minimum wage, the likelihood contribution captures the probability that a worker's wage is defined by the Nash bargaining framework, and not by utilizing the minimum wage floor. In this case, the wage reflects the worker's productivity, her bargaining power in how to split the employment surplus, and the overall market conditions.

We then combine these three likelihood contributions from workers who are unemployed, earn wages equal to the minimum wage, and those who have wages above the minimum wage into a single likelihood function. We use a maximum likelihood estimation strategy to find the set of model parameters—the job arrival rate (ψ), job separation rate (δ), workers' bargaining power (γ), mean of productivity distribution (μ_θ), standard deviation of productivity distribution (σ_θ), and reservation utility (θ^*)—that best fit the observed data.

In our analysis, we account for unobserved heterogeneity by including fixed effects for years and types of crops and tasks that workers are involved in during the NAWS interview.¹⁴ Additionally, we control for variables such as worker demographics (age, gender, citizenship status), job characteristics (hiring process, methods of payment, and experience). Further, we interact our key model parameters (productivity and bargaining power) across employer and worker attributes to understand how the estimates vary across different worker groups. This approach allows us to rigorously examine how different workers contribute to productivity (thus increasing the total size of the pie) and how they share the surplus (allocating the total pie). We discuss our results in the section Results.

Finally, we test whether high profitability and high wages are mutually exclusive for agricultural labor markets, which is not the case in Autor et al. (2020) due to “superstar-firms” argument. That is, we examine if the summary evidence—a positive correlation of wages and profits in a sample of US food retailers and wholesalers—in Figure 1 also holds for agricultural firms. To this end, we generate simulated values for productivity, bargaining power, and surplus across all observations in our data, and then explore the relationship between wages and employer's share of surplus through a summary regression. That is, we use parameters estimated from our preferred structural model specification—one where we interact both productivity and bargaining power with different attributes—and calculate implied values for each element across all observations. Since both productivity and bargaining power vary along heterogeneous attributes, we compute a range of (employer) surplus values as: $(\exp(\mu_\theta + \sigma_\theta^2/2) - \theta^*)(1 - \gamma)$. Finally, we examine the surplus-wage relationship by conducting a logarithmic regression of these calculated surplus values against observed wages. We summarize our findings in Figure 8.

RESULTS

In this section, we present our results from estimating the model in Appendix B. We describe our results in detail below, but the summary is that while we find workers' bargaining power to be roughly 24%, we also document a robust positive elasticity of employer share of surplus and observed wages. This suggests that a view of bargaining power as a sufficient parameter to gauge worker exploitation can be misleading. Instead, we find employers in agriculture, and perhaps the food system more generally, earn more surplus when they pay higher wages after we adjust for heterogeneity in bargaining power and productivity across different worker attributes.

Before we explain the results for the superstar-firm hypothesis, we present the results about bargaining power and workers' productivity as they form a basis for estimation of the surplus-wage elasticity. Table 3 provides the estimates of structural parameters in the log-likelihood function (B.8) in the setting of crop workers in California from 1989 to 2022. Model 1 offers a baseline assessment that unemployed workers regularly face new job openings and have a modest bargaining power, getting about 23.5% of the employment surplus—the gap between worker's productivity and reservation wage. In Model 2, we interact the bargaining parameter with several worker and employer attributes and find that factors such as citizenship, gender, and farm work experience influence bargaining power.

Model 1 in Table 3 controls for year, crop, and task fixed effects, and for worker attributes: age, age-squared, gender, education, years of farm work, foreign-born, and citizenship status. It shows the job arrival rate (ψ) is 0.118, which means there is 11.8% probability that an

TABLE 3 Estimates of worker bargaining power in California, 1989–2022.

Parameter/variable	Notation	Model 1		Model 2	
		Est.	Std. err.	Est.	Std. err.
Job arrival rate	ψ	0.118	0.002	0.116	0.002
Job separation rate	δ	0.180	0.004	0.180	0.004
Mean productivity	μ_θ	2.027	0.006	2.021	0.006
Std. dev. productivity	σ_θ	0.192	0.010	0.187	0.010
Reservation utility	ϑ^*	3.232	0.006	3.225	0.005
Workers' bargaining power	γ	0.235	0.001	0.244	0.003
Heterogeneity of γ					
Citizenship status				−0.016	0.001
Gender				−0.003	0.001
FLC				0.003	0.001
Age				−0.016	0.007
Foreign born				0.019	0.002
Piece rate				−0.023	0.001
Years farm work				−0.100	0.007
Controls					
Year fixed effects		Yes		Yes	
Crop fixed effects		Yes		Yes	
Task fixed effects		Yes		Yes	
Worker attributes		Yes		Yes	
LLF		−74,703.51		−74,159.7	
AIC/N		6.02		5.98	
Number of observations		24,827		24,827	

Note: All estimates obtained with structural search, match, and bargaining model similar to Flinn (2006). Model 1 represents the base model with no heterogeneity in bargaining. In Model 2, we include various worker and employer attributes—worker's citizenship status, gender, Farm Labor Contractor (FLC), age, foreign, piece-rate, and years of farm work experience—in a single model in order to estimate the partial effect of each. Status = 1 if a worker is undocumented. Gender = 1 for males. FLC = 1 if worker is hired by a farm labor contractor, and is 0 if hired directly by a grower. Foreign = 1 for foreign-born workers. Piece = 1 if a worker is paid on a piece-rate basis, and 0 if paid hourly. We include year, crop, and task fixed effects, and control for following worker attributes: age, age-squared, gender, education, years of farm work, foreign-born, and citizenship status.

Source: The data source is NAWS, US Department of Labor for the period 1989–2022 (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

unemployed worker will get a job offer in a year. The job separation rate (δ) is 0.180, reflecting that an employed worker has 18.0% chance of losing her job in a given year. Worker productivity has a lognormal distribution with parameters $\mu_\theta = 2.027$ and $\sigma_\theta = 0.192$. Using the expression of match-value, $\exp(\mu_\theta + \sigma_\theta^2/2)$, this translates to a mean productivity of \$7.734 per hour. Meanwhile, worker's reservation utility (ϑ^*) is \$3.232 per hour. As mean productivity is higher than reservation utility, our estimates imply an employment surplus of \$4.502 per hour. Workers and employers share this surplus based on workers' bargaining power parameter,

$\gamma = 0.235$. This suggests that workers take an average of 23.5% of the surplus, and employers (FLC or growers) capture the remaining share.

Model 2 builds on Model 1 by interacting the bargaining power parameter with worker and firm attributes: citizenship status (= 1 if a worker is undocumented), gender (= 1 for males), Farm Labor Contractor (FLC = 1 if hired by a contractor), age, foreign born (= 1 for foreign born workers), and piece (= 1 if paid on a piece-rate basis). The purpose here is to understand how bargaining power varies by different labor-related characteristics. After the interaction, structural parameters take slightly different values. Workers' bargaining power is 24.4%, and it differs considerably across different worker groups. We find that workers who are undocumented, males, older, earn piece-rate, and with more farm experience have a lower bargaining power than their counterparts, which we hypothesize as resulting from vulnerability, limited mobility, and lack of legal protections for such worker groups. On the other hand, our analysis shows that workers who are hired by farm labor contractors and are foreign-born tend to have a higher bargaining power, a result likely driven by collective leverage, contractor negotiations, and stronger labor networks. We visually depict this heterogeneity of bargaining power in Figure 6.

One objective in this study is to understand not just bargaining power but also find how workers' productivity varies by worker and employer characteristics. To this end, we interact mean productivity with worker and employer attributes as above. Table 4 reports the findings. Model 1 has the same results as Model 1 in Table 3 because we do not change the baseline specification. Model 2 shows findings after the interaction. We notice that an increase in job creation rate ($\psi = 13.8\%$), separation rate ($\delta = 19.3\%$), mean productivity ($\mu_g = 2.132$), standard

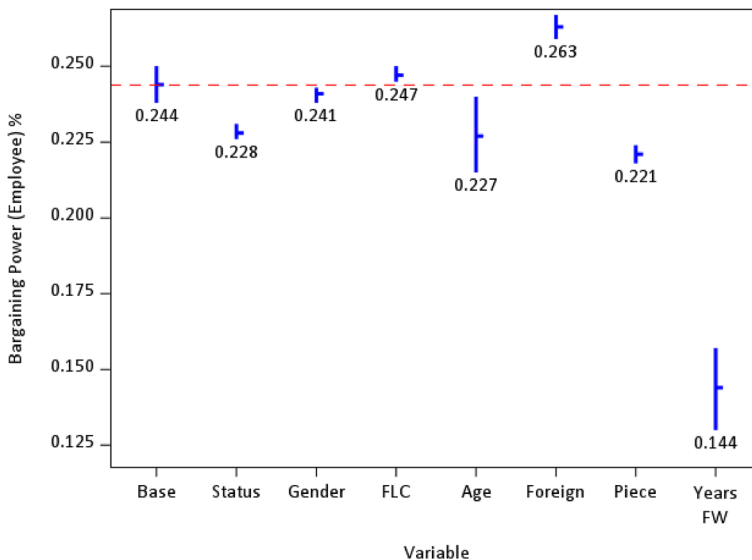


FIGURE 6 Heterogeneity in worker bargaining power in California, 1989–2022. The figure shows how workers' bargaining power differs across worker and employer attributes: worker's citizenship status, gender, Farm Labor Contractor (FLC), age, foreign, piece-rate, and years of farm work experience. Status = 1 if a worker is undocumented. Gender = 1 for males. FLC = 1 if worker is hired by a farm labor contractor, and is 0 if hired directly by a grower. Foreign = 1 for foreign-born workers. Piece = 1 if a worker is paid on a piece-rate basis, and 0 if paid hourly. *Source:* The data source is NAWS, US Department of Labor (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

TABLE 4 Estimates of worker productivity in California, 1989–2022.

Parameter/variable	Notation	Model 1		Model 2	
		Est.	Std. err.	Est.	Std. err.
Job arrival rate	ψ	0.118	0.002	0.138	0.002
Job separation rate	δ	0.180	0.004	0.193	0.004
Mean productivity	μ_θ	2.027	0.006	2.132	0.019
Std. dev. productivity	σ_θ	0.192	0.010	0.273	0.012
Reservation utility	ϑ^*	3.232	0.006	3.234	0.006
Workers' bargaining power	γ	0.235	0.001	0.234	0.001
Heterogeneity of μ_θ					
Citizenship status				−0.165	0.014
Gender				0.168	0.010
FLC				0.002	0.009
Age				−0.180	0.055
Foreign born				−0.055	0.014
Piece rate				0.064	0.012
Years farm work				0.272	0.062
Controls					
Year fixed effects		Yes		Yes	
Crop fixed effects		Yes		Yes	
Task fixed effects		Yes		Yes	
Worker attributes		Yes		Yes	
LLF		−74,703.51		−74,014.55	
AIC/N		6.02		5.97	
Number of observations		24,827		24,827	

Note: All estimates obtained with structural search, match, and bargaining model similar to Flinn (2006). Model 1 represents the base model with no heterogeneity in productivity. In Model 2, we include various worker and employer attributes—worker's citizenship status, gender, Farm Labor Contractor (FLC), age, foreign, piece-rate, and years of farm work experience—in a single model in order to estimate the partial effect of each. Status = 1 if a worker is undocumented. Gender = 1 for males. FLC = 1 if worker is hired by a farm labor contractor, and is 0 if hired directly by a grower. Foreign = 1 for foreign-born workers. Piece = 1 if a worker is paid on a piece-rate basis, and 0 if paid hourly. We include year, crop, and task fixed effects, and control for following worker attributes: age, age-squared, gender, education, years of farm work, foreign-born, and citizenship status.

Source: The data source is NAWS, US Department of Labor for the period 1989–2022 (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

deviation of productivity ($\sigma_\theta = 0.273$), and reservation utility ($\vartheta^* = 3.234$), and a slight decrease in bargaining power ($\gamma = 0.234$), relative to the values in Model 1. As Model 2 has slightly higher LLF value and lower AIC value, we argue it indicates a better fit of our data than Model 1. Further, we interpret Model 2 results as correcting for omission of differential impacts of worker attributes. Applying the expression of match-value, $\exp(\mu_\theta + \sigma_\theta^2/2)$, gives workers' mean productivity as \$8.756 per hour. A heterogeneity analysis shows that μ_θ is lower among workers who are undocumented, older, and are foreign born, which is likely due to limited formal

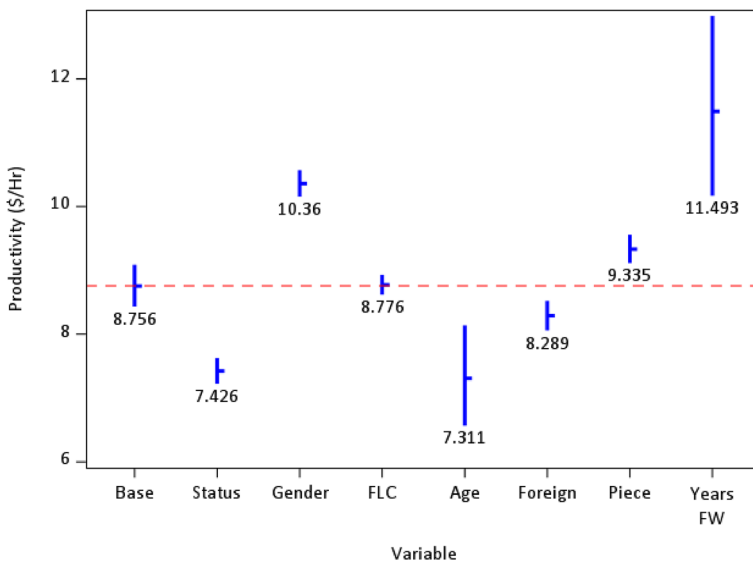


FIGURE 7 Heterogeneity in worker productivity in California, 1989–2022. The figure shows how workers' productivity differs across worker and employer attributes: worker's citizenship status, gender, Farm Labor Contractor (FLC), age, foreign, piece-rate, and years of farm work experience. Status = 1 if a worker is undocumented. Gender = 1 for males. FLC = 1 if worker is hired by a farm labor contractor, and is 0 if hired directly by a grower. Foreign = 1 for foreign-born workers. Piece = 1 if a worker is paid on a piece-rate basis, and 0 if paid hourly. *Source:* The data source is NAWS, US Department of Labor (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

training, language barriers, or physical limitations. Similarly, we find it is higher among workers who are males, earn piece-rate, and with more farm experience, perhaps due to stronger physical abilities, alignment of incentives with output, and accumulated experience.¹⁵ We show the precise estimates of productivity for these groups in Figure 7.

Our next goal ties with Nash bargaining framework where the objective is to model how different agents contribute to “size of the pie” and not just “share the pie.” That is, we want to understand how different worker groups share the employment surplus *relative to* their contribution to productivity. This informs whether workers' share of surplus is solely due to bargaining power effects or whether it aligns with their productivity levels.

To understand how surplus size relates to its distribution, we interact both mean productivity and bargaining parameters with different worker and employer attributes in the same baseline model discussed in Tables 3 and 4. Table 5 reports the findings. Compared to the baseline results in Model 1 in Table 3, we document higher values for job creation rate is (13.9%), job separation rate (19.3%), mean productivity (2.12), standard deviation of productivity (0.281), and workers' bargaining power (24.2%), and a lower value for reservation utility (3.226). Applying the same expression for match-value as above, we find that mean productivity is \$8.67 per hour. In the heterogeneity analysis, we find undocumented workers have lower productivity and lower bargaining power. Such workers likely have lower bargaining power because they are more likely to be in “take it or leave it” (or job-posting in the terminology of bargaining models) or are subject to jobs that are pre-arranged by their contractors. At the same time, worker productivity may be lower due to the likelihood that undocumented workers have less

TABLE 5 Estimates of worker productivity and bargaining in California, 1989–2022.

Variable/parameter	Notation	Model 1		Model 1	
		Est.	Std. err.	Est.	Std. err.
Job arrival rate	ψ	0.139	0.002	0.139	0.002
Job separation rate	δ	0.193	0.004	0.193	0.004
Mean productivity	μ_θ	2.120	0.020	2.120	0.020
Std. dev. productivity	σ_θ	0.281	0.013	0.281	0.013
Reservation utility	ϑ^*	3.226	0.005	3.226	0.005
Workers' bargaining power	γ	0.242	0.003	0.242	0.003
Heterogeneity		Heterogeneity of μ_θ		Heterogeneity of γ	
Citizenship status		−0.155	0.014	−0.013	0.001
Gender		0.178	0.011	−0.007	0.001
FLC		0.000	0.009	0.002	0.001
Age		−0.169	0.057	−0.015	0.007
Foreign born		−0.066	0.015	0.020	0.002
Piece rate		0.081	0.013	−0.022	0.001
Years farm work		0.366	0.064	−0.099	0.007
Controls					
Year fixed effects		Yes		Yes	
Crop fixed effects		Yes		Yes	
Task fixed effects		Yes		Yes	
Worker attributes		Yes		Yes	
LLF		−73,451.17		−73,451.17	
AIC/N		5.92		5.92	
Number of observations		24,827		24,827	

Note: All estimates obtained with structural search, match, and bargaining model similar to Flinn (2006). The top part of the table represents baseline values for our structural parameters: $\psi, \delta, \mu_\theta, \sigma_\theta, \vartheta^*$ and γ . In the middle part, we include various worker and employer attributes in order to estimate the partial effect of each on both productivity and bargaining power. So, the middle part represents heterogeneity in productivity and bargaining power along different worker groups. We include various worker and employer attributes—worker's citizenship status, gender, Farm Labor Contractor (FLC), age, foreign, piece-rate, and years of farm work experience—in a single model in order to estimate the partial effect of each. Status = 1 if a worker is undocumented. Gender = 1 for males. FLC = 1 if worker is hired by a farm labor contractor, and is 0 if hired directly by a grower. Foreign = 1 for foreign-born workers. Piece = 1 if a worker is paid on a piece-rate basis, and 0 if paid hourly. We include year, crop, and task fixed effects, and control for following worker attributes: age, age-squared, gender, education, years of farm work, foreign-born, and citizenship status.

Source: The data source is NAWS, US Department of Labor for the period 1989–2022 (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

training, are more likely to be migrant workers, or are otherwise more temporary and less committed to a particular workplace.

In Table 5, we also find that males, piece-rate earners, and workers with more years of farm work experience tend to have higher productivity but lower bargaining power. One possible rationale for piece-rate earners and experienced workers to have higher productivity is their familiarity with tasks and direct alignment of incentives with output. We also find that workers

hired by contractors have the same productivity as those hired by growers, but the FLC workers have higher bargaining power, likely due to collective leverage, a result similar to Card et al. (2004) who find workers with union membership have higher bargaining power. Finally, we find workers who are older and foreign born have lower productivity, possibly due to physical limitations and language barriers, but have higher bargaining power, as they likely bring in diverse experiences and specialized skills. We summarize this heterogeneity in productivity and bargaining power in Table 6.

We exploit the heterogeneity in bargaining power and productivity to examine how observed wages relate to employers' share of surplus. We illustrate our findings in Figure 8. The graphs shows that higher wages are correlated with higher employer surplus, with surplus-wage elasticity $\epsilon = 0.1619$ and t-ratio 29.54. This aligns with both our summary evidence from food retailing (Figure 1) and Autor et al. (2020) who argue that wages and profitability are not always mutually exclusive for superstar firms. In our case, it means that agricultural employers can gain more surplus by compensating their workers with higher wages, and not always via traditional explanations of exploitation and "winner-take-all" approach.

In summary, we document the importance of looking at productivity effects, in addition to bargaining effects as done in classical studies of imperfect competition. Our analysis highlights how an employer–employee match both increases the size of the pie and examines how the pie is distributed among employers and workers. Specifically, we argue workers who have a lower bargaining power can still benefit by having higher productivity. While we do not have enough information in the NAWS data to make a definitive statement, it is possible that productivity effects offset the bargaining effects among workers who are male, piece-rate earners, and with more farm experience. If this is a case, such workers are still better-off despite having a lower bargaining power. Further, we document a robust positive elasticity of employer surplus with wages, suggesting that agricultural firms can gain more by paying higher wages to their workers. This explains how bargaining equilibria eliminates the notion of "zero sum game" in traditional models of imperfect competition.

Our central results on bargaining equilibria and surplus-wage elasticity are that productive and profitable firms cannot always be characterized as those who suppress wages or "exploit workers," and workers with low bargaining power can still be better off by generating higher

TABLE 6 Changes in mean productivity and bargaining power by worker attributes.

Worker attributes	Direction of change	
	Mean productivity (μ_p)	Bargaining power (γ)
Undocumented	↓	↓
Male	↑	↓
Under FLC	No change	↑
Older	↓	↑
Foreign born	↓	↑
Piece rate	↑	↓
More farm experience	↑	↓

Note: The table shows how workers' mean productivity and bargaining power vary across different worker attributes. Arrows indicate the increase (↑) or decrease (↓) in the respective attribute, while "No Change" signifies no significant change, relative to the corresponding counterparts of worker attributes.

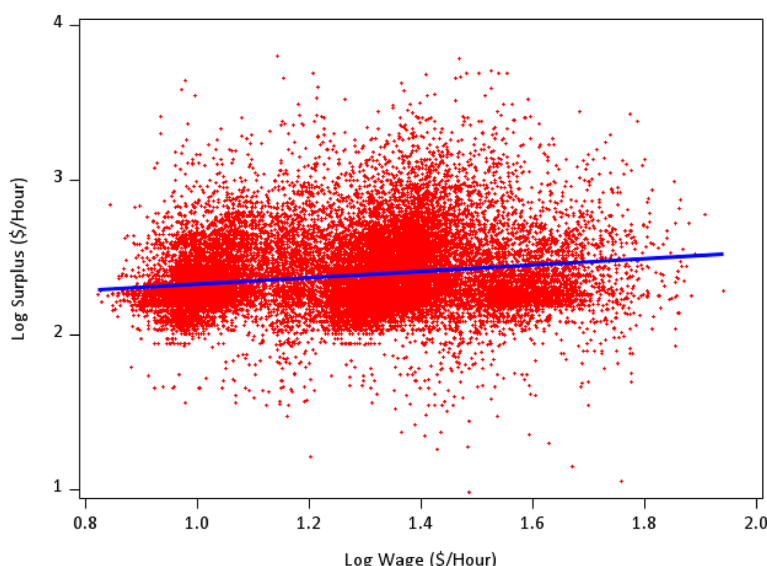


FIGURE 8 Relationship between wages and employment surplus for CA farms. The figure shows how the log of employment surplus relates to log of wages for California crop workers. Surplus-wage elasticity is 0.16192 and the t-ratio is 29.54. *Source:* The data source is NAWS, US Department of Labor (<https://www.dol.gov/agencies/eta/national-agricultural-workers-survey>).

productivity for their employer. These findings do not align with usual policy interventions such as POTUS (2021) that maintain imperfect competition in labor markets leads to worker exploitation. Similarly, we challenge the narratives in popular media and public debates that productive and high-performing food-system employers necessarily win by making their workers lose. We argue that this zero-sum game does not hold, at least in the setting of US crop employers and workers. Future policy designs for the food sector should consider the possibility that a worker with a low negotiation power can still be better off by contributing higher value to her employer, and that employers can provide better compensation as they become more profitable.

CONCLUSIONS AND IMPLICATIONS

In this paper, we examine the underlying tenets of the superstar-firm theory in an agricultural setting. We interpret the superstar effects as implying that firms can simultaneously earn higher profits and pay better wages to their workers. We document that agricultural firms gain more surplus by paying higher wages, a result similar to Autor et al. (2020). This is consistent with empirical findings that larger agricultural farms in the US maintain better farm profitability relative to smaller farms.¹⁶ Our findings are in sharp contrast to more usual models of labor market power that imply worker exploitation and present imperfect competition as a zero-sum game, where workers lose when employers gain.

We test our hypothesis using a structural model of search, match, and bargaining from the Diamond–Mortensen–Pissarides tradition to assess the heterogeneity in bargaining power and productivity of crop workers using NAWS data for crop workers in California. First, the results

from model estimation show that workers, on average, have a bargaining power parameter of 0.242, which implies they receive less than one-fourth of the employment surplus and their employers (growers or contractors) take the remaining share. Our heterogeneity analysis suggests that bargaining power differs across worker groups: Foreign-born workers and those hired by contractors have slightly higher bargaining power, while females, older workers, piece-rate earners, and more experienced farm workers have lower bargaining power relative to their corresponding counterparts.

Our study also suggests that the usual approach of labor market power, based solely on bargaining power effects, as a direct implication of exploitation may be oversimplified. Instead, we propose to focus on whether there is an alignment between workers' productivity and their share of the employment surplus. For instance, some workers can have high productivity but low bargaining power and can still manage to bargain for better offers due to the relative strength of productivity effects over bargaining effects. We examine how variables such as age, gender, experience, mode of payment, and immigration status affect both productivity and bargaining power. In doing so, we are able to understand which worker groups are likely under-compensated relative to their productivity. We argue that this approach improves theoretical understanding of labor market dynamics but can also reveal practical insights for policy-making. Our findings indicate that if employers recognize or are convinced that paying higher wages benefits their operations, they have the economic incentive to increase worker compensation. Further, our results add to recent antitrust efforts to mitigate monopsony power in labor markets, but argue that productive and profitable firms need not always be the ones that exploit workers or underpay the workers.

ENDNOTES

- ¹ Autor et al. (2020) use this idea to reconcile observations of a generally-declining labor share of value with increasing product-market concentration and relatively high compensation by successful firms.
- ² See the USDA-NASS report from 2013 on how farm profitability varies with farm size: <https://www.ers.usda.gov/data-products/chart-gallery/gallery/chart-detail?chartId=78328>.
- ³ In our study period 1989–2022, crop workers had an average age of 35 years, and 40% of workers in 2022 were over 47 years, both indicating an aging workforce (authors' calculation based on the NAWS data from <https://www.dol.gov/agencies/eta/national-agricultural-workers-survey/data>).
- ⁴ In 2017, California lost nearly 40% of its farm workers between 2002 and 2014 according to <https://www.economist.com/united-states/2017/07/27/if-america-is-overrun-by-low-skilled-migrants> and <https://www.nass.usda.gov/AgCensus/>.
- ⁵ Note that our data does not describe individual firms, but search, match, and bargaining models capture equilibrium outcomes so firm choices are implicit in the deals accepted by workers.
- ⁶ The public-version of NAWS data is available for the period 1989–2020 here: <https://www.dol.gov/agencies/eta/national-agricultural-workers-survey/data>.
- ⁷ This means the workers in our final sample work in relatively-similar industries for employers that sell into similar output markets. Further, how state-level minimum wage policy applies to agricultural workers is generally unique to each state, and focusing on California avoids incorporating cross-state minimum wages (<https://nationalaglawcenter.org/state-compilations/agpay/minimumwage/>).
- ⁸ This also aligns well with the concept of Nash bargaining equilibrium where the idea is to increase the “size of the pie,” not just “share of the pie,” so even the side with less bargaining power ends up with more surplus than would have been the case otherwise.
- ⁹ We also note that this result on surplus-wage elasticity likely applies to broader food system as we show in a summary evidence that a sample of more profitable retail and wholesale firms actually pay higher wages

(Figure 1), which is intuitive, but completely at odds with the traditional notion of large firms exercising exploitative market power over workers.

- ¹⁰ See Azar and Marinescu (2024) for a review on labor monopsony literature.
- ¹¹ Bargaining power parameter is a statistic which summarizes “labor market position” of a firm or a worker (Flinn, 2006). For example, low-skilled workers for whom there are many substitutes may have lower bargaining power than high-skilled workers with smaller number of substitutes.
- ¹² This does not imply that employers and workers literally negotiate each term of their employment contract one-on-one; rather, it means every labor market produces outcomes that resemble the optimal search behavior of the two parties (Burdett & Mortensen, 1998; Flinn, 2006).
- ¹³ Full list of NAWs 12 regions and states in each region is available at: https://www.dol.gov/sites/dolgov/files/ETA/naws/pdfs/Map_of_Naws_17_Sampling_Regions.pdf.
- ¹⁴ Although we cannot include county fixed effects due to data limitations, this omission does not affect our core findings. Our model relies on crop- and task-specific fixed effects that likely capture much of the variation coming from labor shortages and inter-county migration. Further, we model employer–employee equilibrium, so any border-related wage suppression is likely to be understood by workers, who will move away from border regions as a result and equalize the any bargaining power differentials that may arise.
- ¹⁵ While our findings suggest that piece-rate workers appear to have lower bargaining power but higher productivity, this may be influenced by a “fixed-rate bias.” In practice, employers often set uniform piece rates across workers to maintain fairness and share surplus, rather than adjusting rates based on individual worker’s productivity. Ideally, a true efficiency wage model would offer higher rates to more productive workers, but fairness concerns and practical considerations lead to fixed rates that mask the underlying dynamics of bargaining power and productivity.
- ¹⁶ See USDA-NASS report from 2013 on how farm profitability varies with farm size: <https://www.ers.usda.gov/data-products/chart-gallery/gallery/chart-detail?chartId=78328>.
- ¹⁷ While our model assumes immediate match productivity observability, this should be interpreted in an “as if” sense—the observed wage and unemployment duration patterns reflect outcomes that would occur if match values were immediately known.
- ¹⁸ The assumption of lognormal distribution of the match-value θ holds true empirically in wage and income data (Flinn, 2006).

REFERENCES

- AgAmerica. 2022. “Three Ways Immigration Reform Could Help U.S. Farmers.” <https://agamerica.com/blog/farm-labor-shortage/>.
- Arnold, D. 2021. “Mergers and Acquisitions, Local Labor Market Concentration, and Worker Outcomes.” Working paper, Department of Economics, Princeton University.
- Autor, D., D. Dorn, L. F. Katz, C. Patterson, and J. Van Reenen. 2020. “The Fall of the Labor Share and the Rise of Superstar Firms.” *Quarterly Journal of Economics* 135(2): 645–709.
- Azar, José A., Steven T. Berry, and Ioana Marinescu. 2022. “Estimating Labor Market Power.” National Bureau of Economic Research Working Paper 30365.
- Azar, José, and Ioana Marinescu. 2024. “Monopsony Power in the Labor Market: From Theory to Policy.” *Annual Review of Economics* 16: 491–518.
- Berger, D., K. Herkenhoff, and S. Mongey. 2022. “Labor Market Power.” *American Economic Review* 112(4): 1147–93.
- Berger, D. W., K. F. Herkenhoff, A. R. Kostol, and S. Mongey. 2023. “An Anatomy of Monopsony: Search Frictions, Amenities and Bargaining in Concentrated Markets.” National Bureau of Economic Research, No. w31149.
- Bhaskar, V., Alan Manning, and Ted To. 2002. “Oligopsony and Monopsonistic Competition in Labor Markets.” *Journal of Economic Perspectives* 16(2): 155–174. <https://doi.org/10.1257/0895330027300>.

- Blau, Francine D., and Lawrence M. Kahn. 2017. "The Gender Wage Gap: Extent, Trends, and Explanations." *Journal of Economic Literature* 55(3): 789–865.
- Burdett, Kenneth, and Dale T. Mortensen. 1998. "Wage Differentials, Employer Size, and Unemployment." *International Economic Review* 39(2): 257–273. <https://doi.org/10.2307/2527292>.
- Cahuc, P., F. Postel-Vinay, and J. M. Robin. 2006. "Wage Bargaining With On-The-Job Search: Theory and Evidence." *Econometrica* 74(2): 323–364.
- Card, David. 2022. "Who Set Your Wage?" *American Economic Review* 112(4): 1075–90.
- Card, David, Ana Rute Cardoso, Joerg Heining, and Patrick M. Kline. 2021. "Firms and Labor Market Inequality: Evidence and Some Theory." *Journal of Labor Economics* 36(S1): S13–S70.
- Card, David, Thomas Lemieux, and W. Craig Riddell. 2004. "Unions and Wage Inequality." *Journal of Labor Research* 25(4): 519–559. <https://doi.org/10.1007/s12122-004-1011-z>.
- Dey, M. S., and C. J. Flinn. 2005. "An Equilibrium Model of Health Insurance Provision and Wage Determination." *Econometrica* 73(2): 571–627.
- Diamond, Peter A. 1982. "Wage Determination and Efficiency in Search Equilibrium." *The Review of Economic Studies* 49(2): 217–227. <https://doi.org/10.2307/2297271>.
- Faberman, R. Jason, Andreas I. Mueller, and Ayşegül Şahin. 2022. "Has the Willingness to Work Fallen During the Covid Pandemic?" National Bureau of Economic Research Working Paper.
- Fan, Mingyue, Susan Gabbard, Anita A. Pena, and Jeffrey M. Perloff. 2015. "Why Do Fewer Agricultural Workers Migrate Now?" *American Journal of Agricultural Economics* 97(3): 665–679. <https://doi.org/10.1093/ajae/aau115>.
- Federal Trade Commission. 2023. "FTC, Department of Labor Partner to Protect Workers From Anticompetitive, Unfair, and Deceptive Practices." Press Release. <https://www.ftc.gov/news-events/news/press-releases/2023/09/ftc-department-labor-partner-protect-workers-anticompetitive-unfair-deceptive-practices>.
- Flinn, C. J. 2006. "Minimum Wage Effects on Labor Market Outcomes Under Search, Matching, and Endogenous Contact Rates." *Econometrica* 74(4): 1013–62.
- Flinn, Christopher, and James Mabli. 2008. "On-The-Job Search, Minimum Wages, and Labor Market Outcomes in an Equilibrium Bargaining Framework." Collegio Carlo Alberto. Carlo Alberto Notebooks 91.
- Goebel, Christian, and Thomas Zwick. 2009. "Age and Productivity—Evidence From Linked Employer Employee Data." ZEW—Centre for European Economic Research Discussion Paper 09-020.
- Gold, Amanda, Wenson Fung, Susan Gabbard, and Daniel Carroll. 2019. "Findings From the National Agricultural Workers Survey (NAWS) 2019–2020: A Demographic and Employment Profile of United States Farmworkers." <https://www.dol.gov/sites/dolgov/files/ETA/naaws/pdfs/NAWS%20Research%20%Report%2016.pdf>.
- Gottfries, Axel, and Gregor Jarosch. 2023. "Dynamic Monopsony With Large Firms and Noncompetes." National Bureau of Economic Research Working Paper 31965. Working Paper.
- Jarosch, Gregor, Jan Sebastian Nimczik, and Isaac Sorkin. 2024. "Granular Search, Market Structure, and Wages." *Review of Economic Studies* 91: 3569–3607.
- Kandilov, Amy M. G., and Ivan T. Kandilov. 2010. "The Effect of Legalization on Wages and Health Insurance: Evidence From the National Agricultural Workers Survey." *Applied Economic Perspectives and Policy* 32(4): 604–623.
- Katz, Lawrence F. 1986. "Efficiency Wage Theories: A Partial Evaluation." In *NBER Macroeconomics Annual 1986*, Vol 1, edited by Stanley Fischer, 235–290. Cambridge, MA: MIT Press.
- Lazear, Edward P. 1999. "Culture and Language." *Journal of Political Economy* 107(S6): S95–S126.
- Li, Axiang, and Jeffrey J. Reimer. 2021. "The US Market for Agricultural Labor: Evidence From the National Agricultural Workers Survey." *Applied Economic Perspectives and Policy* 43(4): 1125–39.
- Manning, Alan. 2003. *Monopsony in Motion: Imperfect Competition in Labor Markets*. Princeton, NJ: Princeton University Press.
- Mortensen, Dale T. 1982. In *The Economics of Information and Uncertainty*, edited by John J. McCall, 233–258. Chicago: University of Chicago Press.
- Mortensen, Dale T., and Christopher A. Pissarides. 1994. "Job Creation and Job Destruction in the Theory of Unemployment." *The Review of Economic Studies* 61(3): 397–415.
- Naidu, S., and E. A. Posner. 2022. "Labor Monopsony and the Limits of the Law." *Journal of Human Resources* 57(S): S284–S323.

- Nash, John. 1951. "Non-Cooperative Games." *Annals of Mathematics* 54(2): 286–295. <https://doi.org/10.2307/1969529>.
- National Agricultural Workers Survey. 2023. "Statistical Methods of the National Agricultural Workers Survey." <https://rb.gy/om21z8>.
- Pissarides, C. A. 2000. *Equilibrium Unemployment Theory*. Cambridge, MA: MIT Press.
- Postel Vinay, Fabien, and Jean-Marc Robin. 2002. "Equilibrium Wage Dispersion With Worker and Employer Heterogeneity." *Econometrica* 70(6): 2295–2350. <https://doi.org/10.1111/j.1468-0262.2002.00441.x>.
- POTUS. 2021. "Executive Order on Promoting Competition in the American Economy." <https://www.whitehouse.gov/briefing-room/presidential-actions/2021/07/09/executive-order-on-promoting-competition-in-the-american-economy/>.
- Richards, T. J. 2018. "Immigration Reform and Farm Labor Markets." *American Journal of Agricultural Economics* 100: 1050–71.
- Richards, Timothy J. 2020. "Income Targeting and Farm Labor Supply." *American Journal of Agricultural Economics* 102(2): 419–438.
- Richards, Timothy J., and Zachariah Rutledge. 2023. "Food System Labor and Bargaining Power." *Food Policy* 119: 102502.
- Sexton, Richard J. 2013. "Market Power, Misconceptions, and Modern Agricultural Markets." *American Journal of Agricultural Economics* 95(2): 209–219.
- U.S. Department of Agriculture. 2023. "Size and Composition of the U.S. Agricultural Workforce." <https://www.ers.usda.gov/topics/farm-economy/farm-labor/>.
- Vaghul, Kavya, and Ben Zipperer. 2022. "Historical State and Sub-State Minimum Wages." Version 1.4.0. <https://github.com/benzipperer/historicalminwage/releases/tag/v1.4.0>.
- Yeh, C., C. Macaluso, and B. Hershbein. 2022. "Monopsony in the US Labor Market." *American Economic Review* 112(7): 2099–2138.

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APPENDIX A

MODEL DETAILS

In this section, we provide formal details of our structural model. We base the discussion on Pissarides (2000) and Flinn (2006) throughout and extend both to incorporate institutional features of the labor market in the US crop sector.

Environment

Labor market status. The labor market consists of workers—either unemployed and looking for jobs or currently employed under a grower or a Farm Labor Contractor (FLC)—and employers (either growers or FLCs). The job arrival rate is ψ and denotes the probability that a worker gets a job offer in a given year t . The job separation rate is δ which is the probability that an existing job ends (that is, the employed worker becomes unemployed) in t . Both parameters ψ and δ are exogenous. The employers and workers are forward-looking and discount the future at discount rate r .

Search, match, and bargaining. In the NAWS data, we do not observe workers looking for an alternate job while being currently employed. So our model excludes *on-the-job-search* aspect of the labor market. A match between a worker and an employer generates the productivity-value or match-value ϑ . Both sides observe ϑ which is exogenously determined as a function of worker attributes, employer's production technology, and quality of employer–employee match.¹⁷ We assume ϑ to follow a lognormal distribution: $\ln(\vartheta) \sim \mathcal{N}(\mu_\vartheta, \sigma_\vartheta^2)$.¹⁸ Its cumulative distribution function is denoted as $\Phi(\vartheta)$ and probability density function as $\phi(\vartheta)$. The distribution of ϑ between employer and worker is based on worker's bargaining power parameter $\gamma \in (0, 1)$, equivalently employer's bargaining parameter $1 - \gamma$, which is exogenous to both and measures “negotiation skills” determined by worker and market attributes. Intuitively, γ measures the fraction of the employment surplus that the worker receives.

Wage setting mechanism. The determination of a wage w follows a Nash bargaining framework, where worker's bargaining power informs how the productivity-value from an employer–employee match informs the distribution of employment surplus between the worker and the employer. In particular, the worker's wage is a weighted average of the worker's share of the employment surplus, which is determined by their bargaining power γ , and the outside option, such as the reservation wage b , called *worker's disagreement profit*. If the bargaining process results in a wage less than the binding minimum wage $w_{\min}$, then the wage w is raised to the level of $w_{\min}$. Hence, $w_{\min}$ acts as the greatest lower bound in the wage distribution.

Value functions

The model comprises three value functions denoting the utility of being unemployed, employed, and the firm's value of employing a worker.

Unemployed worker's value function (U). An unemployed worker derives a utility b (that is, unemployment benefit) and has a probability ψ of finding a job in each period t . Thus, her utility or the value function from being unemployed is:

$$U = b + \underbrace{\psi \beta \int_{w_{\min}}^{\infty} \max(E(w) - U, 0) d\Phi(w)}_{\text{Expected present value of surplus}}, \quad (\text{A.1})$$

where $\beta = \frac{1}{1+r}$ is the discount factor with discount rate r , and $E(w)$ is the value function for an employed worker earning wage w . Here, the integral sums over the distribution of those wages that make employment more attractive than staying unemployed. The right-hand-side in



Equation (A.1) indicates expected present value of employment-surplus from a job with $w \geq w_{\min}$.

Employed worker's value function ($E(w)$). An employed worker earns a wage w and encounters a job separation probability δ . The worker's value function from being employed is:

$$E(w) = w + \beta[\delta U + (1 - \delta)E(w)], \quad (\text{A.2})$$

where with probability δ , the worker becomes unemployed and receives the utility U . With probability $1 - \delta$, the worker continues to be employed, earning the wage w and getting the value $E(w)$.

We solve the implicit Equation (A.2) for $E(w)$ and get:

$$E(w) = \frac{w + \beta\delta U}{1 - \beta(1 - \delta)}. \quad (\text{A.3})$$

Firm's value function ($J(w)$). We assume that labor is the only means of production for the US crop employers. This means that employer's profit when there is no employer–employee match after a search process—its disagreement outcome—is zero as it does not hire a worker and hence makes no revenue. If the firm gets a match with productivity ϑ and hires the worker by paying wage w , then its profit is $\vartheta - w$ adjusted by the same discount factor as that of the worker:

$$J(w) = \vartheta - w + \beta(1 - \delta)J(w), \quad (\text{A.4})$$

where with probability $1 - \delta$, the worker remains employed, and the firm continues to earn the surplus $\vartheta - w$.

We solve Equation (A.4) for $J(w)$ and get:

$$J(w) = \frac{\vartheta - w}{1 - \beta(1 - \delta)}. \quad (\text{A.5})$$

Wage determination

After an employer–employee match, the two split the employment surplus and bargain for wages following a Nash bargaining framework:

$$w^* = \arg \max_w [E(w) - U]^\gamma \cdot [J(w)]^{1-\gamma}. \quad (\text{A.6})$$

Here, the two sides negotiate for wages based on individual bargaining power parameters. Intuitively, the idea is to increase not just the *share* of the “pie” (the surplus) but also to increase the *size* of the pie.

We substitute employed-worker's and employer's value functions (A.3) and (A.5) into the Nash bargaining Equation (A.6) and get:

$$w^* = \arg \max_w \left[\frac{w + \beta \delta U}{1 - \beta(1 - \delta)} - U \right]^\gamma \cdot \left[\frac{\vartheta - w}{1 - \beta(1 - \delta)} \right]^{1-\gamma}. \quad (\text{A.7})$$

Taking the first-order condition with respect to w , we get the equilibrium wage that solves the Nash bargaining problem (A.7):

$$w^* = \gamma \vartheta + (1 - \gamma) rU, \quad (\text{A.8})$$

where rU is the present value of being unemployed.

Minimum wage constraint. We assume that minimum wage ($w_{\min}$) is in place, and is exogenous and binding to all the employers in our datasets. This means that employers are required to pay at least $w_{\min}$ and cannot compensate workers through other means to avoid this requirement. Further, it implies that employers earn positive profits only when $w_{\min} > U$, that is when the minimum wage exceeds workers' disagreement profit, since workers do not accept job offers in the alternate scenario. Intuitively, an existence of the minimum wage imposes a lower bound on the set of the feasible wages, thus conditioning employers to sacrifice some of their employment surplus from hiring a worker.

If the minimum wage $w_{\min}$ exceeds the equilibrium wage w^* , then the wage distribution will have a mass point at $w_{\min}$. Formally, the constrained Nash bargaining problem under minimum wage restriction is exactly the same as (A.7) but with an additional constraint $w \geq w_{\min}$ in the optimization. The disagreement value to the worker is now a function of $w_{\min}$ instead of w .

Steady-state conditions

For simplicity, we model the steady-state distribution of unemployment and wages. That is, we make two assumptions about employment and wages.

Assumption 1. We assume that the rate at which unemployed workers find jobs equals the rate at which employed workers lose their jobs:

$$u\psi \int_{w_{\min}}^{\infty} \phi(x) dx = \delta(1 - u), \quad (\text{A.9})$$

where u is the unemployment rate, ψ represents the job arrival rate, δ the job separation rate, and ϕ the probability density function of wages. In simpler terms, the left-hand-side represents the product of unemployment rate (share of the workforce that is unemployed), job creation rate (probability that an unemployed worker receives a job offer), and probability that the wage offered is at least the minimum wage ($w_{\min}$). The integral accounts for the fact that workers do not accept all job offers as some offers might be below $w_{\min}$. The right-hand-side represents the rate at the employed workers lose their job and become unemployed.

Assumption 2. Next, we assume that the rate at which unemployed workers find jobs with wage w equals the rate at which employed workers leave jobs with that wage:



$$u\psi\phi(w) = \delta e(w), \quad (\text{A.10})$$

where $e(w)$ is the density of employed workers earning wage w .

In summary, these stationary conditions imply that inflows into unemployment by job destruction balance the outflows into employment by job creation, making unemployment rate constant. They also ensure that the labor market has a stable distribution of wages. Further, as they define the probabilities of observing different labor market conditions such as being unemployed or earning a certain wage, we use them in constructing the likelihood function below.

APPENDIX B

MODEL ESTIMATION AND IDENTIFICATION

In this section, we discuss the estimation and identification of the search, match, and bargaining model developed in Appendix A.

Estimation intuition. Our study employs repeated cross-sectional data on the US crop workers to estimate the parameters of the labor market model in Appendix A. While matched employer–employee data remain ideal for identifying the structural parameters as highlighted in Postel Vinay and Robin (2002) and Cahuc et al. (2006), we follow others in the literature (Flinn, 2006; Richards & Rutledge, 2023) and carry out estimation using the strategies laid out therein.

The estimation involves constructing a likelihood function for each individual who is characterized by two components: the duration of unemployment for individuals seeking employment and the hourly wage rate for those employed. The likelihood function uses this information to categorize individuals into one of the three groups: the unemployed, those earning the minimum wage, and those earning above the minimum wage. Precisely, we calculate the likelihood for the individual to belong to each group, leveraging the probability of being unemployed for a specific duration given the overall unemployment duration distribution within the population.

Likelihood function construction

Our estimation strategy proceeds by constructing likelihood functions for three categories of individuals. To this end, let (t_j, w_j) denote a pair of attributes of each individual in the data: length of unemployment spell in weeks (t_j), which is positive if the person is unemployed, and zero otherwise, and hourly wage (w_j) if the person is employed.

We build the likelihood function by categorizing workers into three groups: unemployed workers, workers earning the minimum wage, and those earning above the minimum wage.

Likelihood for unemployed workers. We assume that unemployment spell follows negative exponential distribution with rate parameter (or hazard rate) $\psi\Phi(w_{\min})$. Here, $\Phi(w_{\min})$ represents the probability that an offered job has wage less than the minimum wage, and the rate parameter denotes how quickly unemployed workers get jobs paying less than $w_{\min}$. Then the conditional probability of observing an unemployment spell of t weeks given the individual is unemployed is:

$$P(t|\text{unemp}) = \psi \Phi(w_{\min}) e^{-\psi \Phi(w_{\min}) t} \quad (\text{B.1})$$

for minimum wage $w_{\min}$ and parameters defined before. The total probability of observing an individual who is unemployed at a given time is:

$$P(\text{unemp}) = \frac{\delta}{\delta + \psi \Phi(w_{\min})}. \quad (\text{B.2})$$

Equation (B.2) denotes the steady-state unemployment rate by balancing the inflow and outflow of unemployment. Given Equations (B.1) and (B.2), the joint probability of observing an individual who is unemployed for t weeks is:

$$P(t, \text{unemp}) = \frac{\delta \psi \Phi(w_{\min}) e^{-\psi \Phi(w_{\min}) t}}{\delta + \psi \Phi(w_{\min})}. \quad (\text{B.3})$$

Likelihood for workers earning minimum wage. The second likelihood function is for an individual who is paid the minimum wage $w_{\min}$. Given that $w_{\min}$ is binding, the probability that the individual is employed and earns wage $w_{\min}$ is:

$$P(w = w_{\min}, \text{emp}) = \frac{\psi \left[\Phi(w_{\min}) - \Phi\left(\frac{w_{\min} - (1-\gamma)rU(w_{\min})}{\gamma}\right) \right]}{\delta + \psi \Phi(w_{\min})}. \quad (\text{B.4})$$

The second expression $\Phi\left(\frac{w_{\min} - (1-\gamma)rU(w_{\min})}{\gamma}\right)$ in the numerator adjusts for the fact that all workers do not naturally earn $w_{\min}$; some would earn less if the minimum wage was not in place. This expression denotes the wages workers would naturally get based on their bargaining power before the minimum wage forces their wages up. The expression for the proportion of workers who would have been paid less than $w_{\min}$ but are now raised to $w_{\min}$ because of the minimum wage law is the difference $\Phi(w_{\min}) - \Phi\left(\frac{w_{\min} - (1-\gamma)rU(w_{\min})}{\gamma}\right)$. Multiplying this difference with the job arrival rate ψ , we get the numerator in Equation (B.4), that is the rate at which job offers which would have been below the minimum wage are rather raised to the minimum wage.

Likelihood for workers earning above minimum wage. For the last component of the likelihood function, we need the share of individuals who earn more than the minimum wage. If an employee's wage exceeds $w_{\min}$, it means that the match-value ϑ is more than $\frac{w_{\min} - (1-\gamma)rU(w_{\min})}{\gamma}$, which is an effective wage offer without a minimum wage in place, taking into account the worker's bargaining power and the value of being unemployed. So, the conditional probability of observing a wage w given employment and that wage exceeds $w_{\min}$ is:

$$p(w|w > w_{\min}, \text{emp}) = \frac{\frac{1}{\gamma} \phi\left(\frac{w - (1-\gamma)rU(w_{\min})}{\gamma}\right)}{\Phi\left(\frac{w_{\min} - (1-\gamma)rU(w_{\min})}{\gamma}\right)}. \quad (\text{B.5})$$



The conditional probability that the wage exceeds $w_{\min}$ given employment is:

$$p(w > w_{\min} | \text{emp}) = \frac{\Phi\left(\frac{w_{\min} - (1-\gamma)rU(w_{\min})}{\gamma}\right)}{\Phi(w_{\min})}. \quad (\text{B.6})$$

Multiplying (B.5) and (B.6), the joint probability that the individual is employed with a wage greater than the minimum wage is:

$$p(w, w > w_{\min}, \text{emp}) = \frac{\frac{\psi}{\gamma} \Phi\left(\frac{w - (1-\gamma)rU(w_{\min})}{\gamma}\right)}{\delta + \psi \Phi(w_{\min})}. \quad (\text{B.7})$$

Aggregate likelihood function for NAWS workers. Let T indicate the count of total individuals in the NAWS data, T_U that of unemployed, T_M that earning the minimum wage, T_A that earning greater than the minimum wage, A the set of workers who earn greater than the minimum wage, B the set of unemployed workers, and $\vartheta^* = rU(w_{\min})$ is the workers' reservation utility. Given this, we combine the likelihood functions for three groups in (B.3), (B.4), and (B.7) and get the log likelihood function for the entire NAWS workers:

$$\begin{aligned} LLF = & \underbrace{T[\ln(\psi) - \ln(\delta + \psi \Phi(w_{\min}))]}_{\text{Total Contribution}} + \underbrace{T_U[\ln(\delta) + \ln \Phi(w_{\min})]}_{\text{Unemp. Workers' Contribution}} \\ & - \underbrace{\psi \Phi(w_{\min}) \sum_{j \in B} t_j}_{\text{Penalty for Unemp. Duration}} + \underbrace{T_M \ln \left(\Phi(w_{\min}) - \Phi\left(\frac{w_{\min} - (1-\gamma)\vartheta^*}{\gamma}\right) \right)}_{\text{Minimum Wage Earners' Contribution}} \\ & - \underbrace{T_A \ln(\gamma) + \sum_{j \in A} \ln \left(\phi\left(\frac{w_j - (1-\gamma)\vartheta^*}{\gamma}\right) \right)}_{\text{Above Minimum Wage Earners' Contribution}}. \end{aligned} \quad (\text{B.8})$$

The goal of estimating this LLF is to find the values of the structural parameters— $\psi, \delta, \gamma, \mu_{\vartheta}, \sigma_{\vartheta}$, and ϑ^* —that maximize the LLF , making the observed data as likely as possible under the model.